

Activity 2 — Team GitHub Documentation

Orders4U Application

Authors	Owen Lindsey & Brennan Bania
Course	CST-323
Instructor	Professor Sluiter
Date	25 January 2026

GitHub Repository Link

Repository URL: <https://github.com/omniV1/CST-323-PairPrograming>

Contributors:

- @omniV1 — Owen Lindsey (Student 2 - Frontend & Design)
 - @BrennanBania — Brennan Bania (Student 1 - Backend)
-

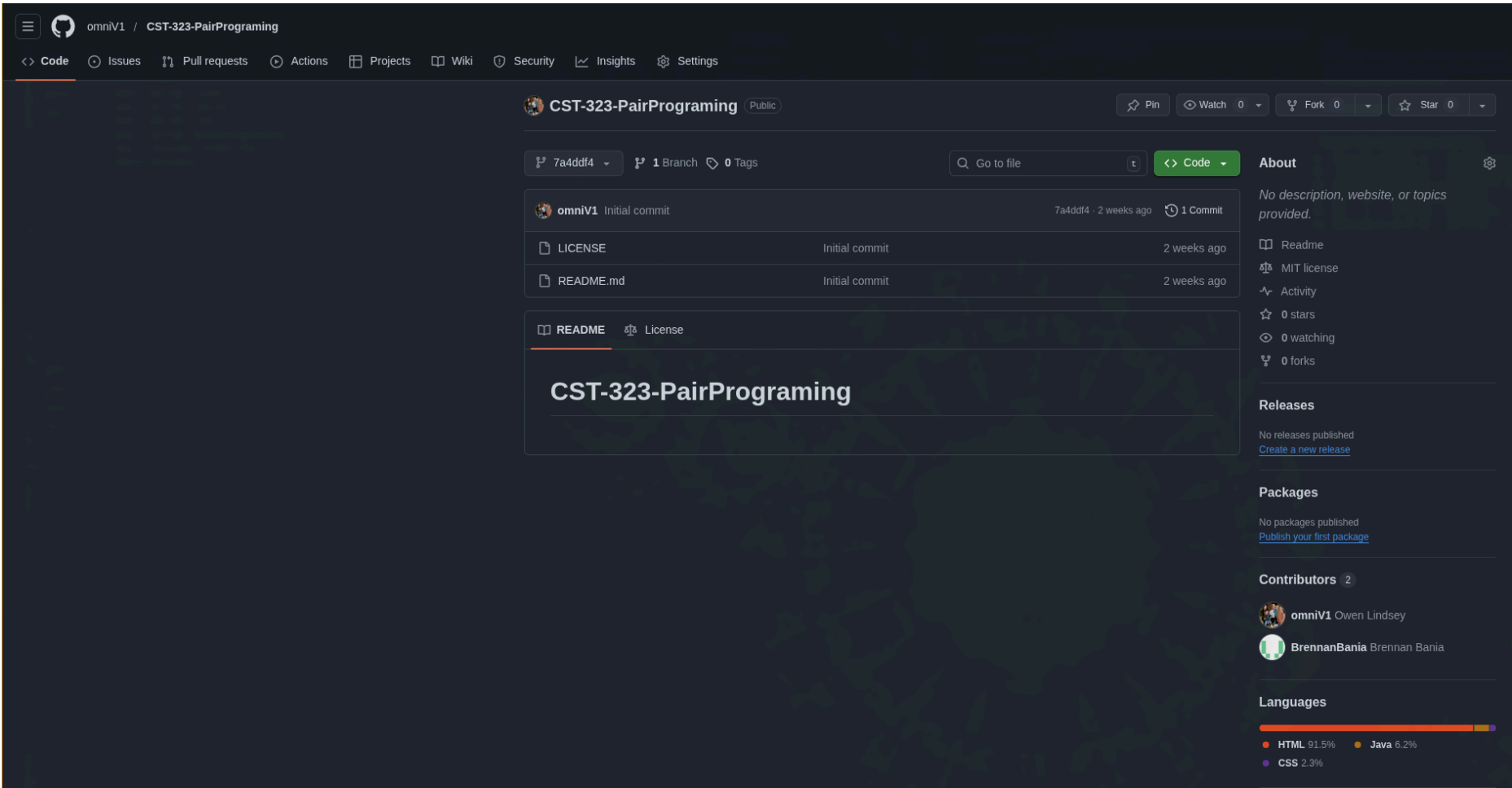
Table of Contents

- [Section 1: Repository Setup Screenshots](#)
 - [Section 2: Branch & Pull Request Workflow Evidence](#)
 - [Section 3: Design Documentation Screenshots \(Student 2\)](#)
 - [Section 4: Backend Development Screenshots \(Student 1\)](#)
 - [Section 5: Frontend Development Screenshots \(Student 2\)](#)
 - [Section 6: Testing & Integration Screenshots](#)
 - [Section 7: Pull Requests](#)
 - [Section 8: AI Exploration Summary](#)
 - [Section 9: Reflection](#)
-

Section 1: Repository Setup Screenshots

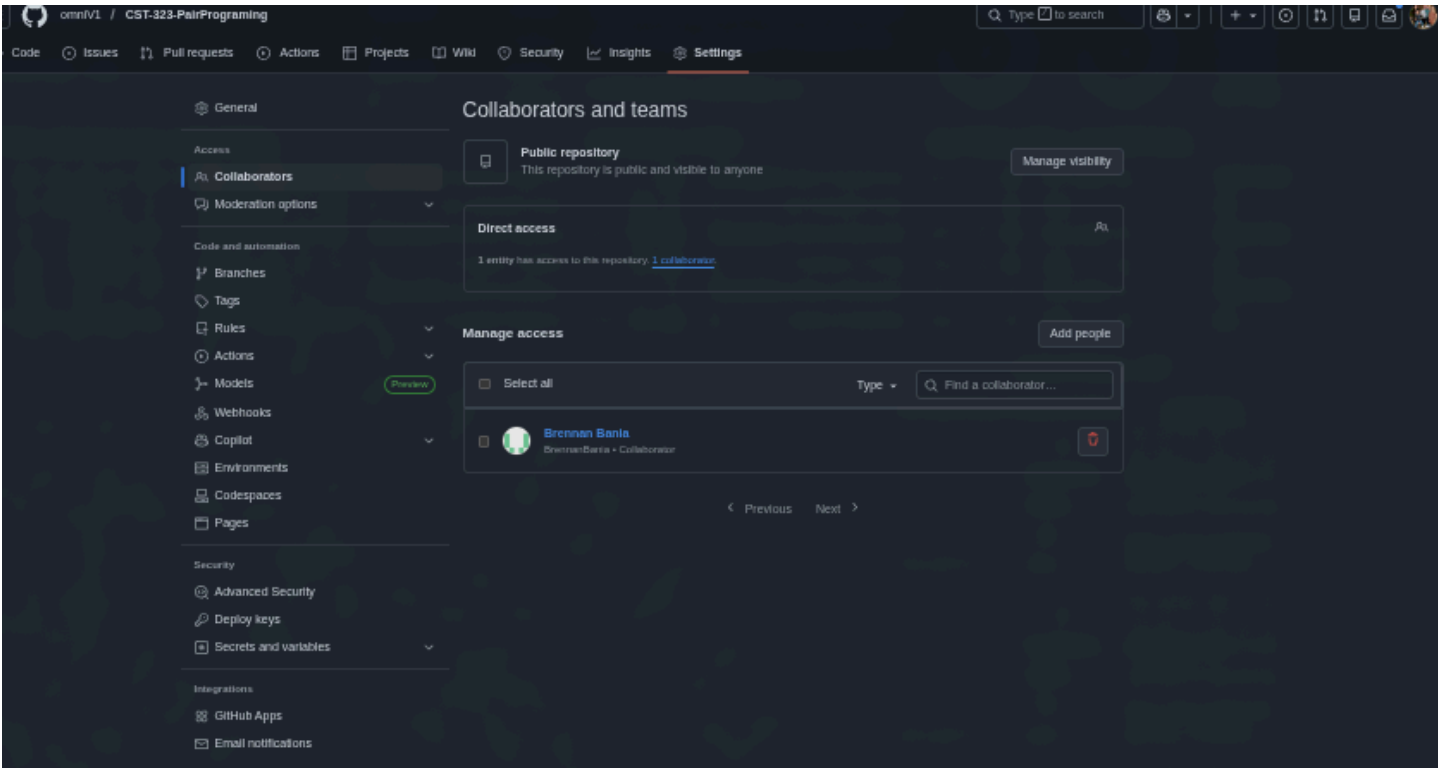
1.1 GitHub Repository Main Page

Caption: Repository main page showing CST-323-PairPrograming as a public repository.



1.2 Collaborators List

Caption: GitHub Settings → Collaborators page showing Brennan Bania (@BrennanBania) added as a collaborator with direct access. The repository owner (@omniV1 - Owen Lindsey) has implicit access and doesn't appear in this list.



Access Summary:

- @omniV1 (Owen Lindsey) — Repository Owner
- @BrennanBania (Brennan Bania) — Collaborator with push access

1.3 Commit History Overview

Caption: Complete commit history showing all commits from both team members.

omniV1 / CST-323-PairProgramming

type to search

Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Commits

main

All users

All time

Commits on Jan 23, 2025

Add UserAdminController with spec-compliant routes and update navigation

BrennanBerna committed 20 hours ago

02c29e0

Merge pull request #11 from omniV1/update-docs

omniV1 authored yesterday

Verified

2cf4b21

Enhance application configuration and documentation. Updated app.json to include PDF export settings. Modified workspace.json to reflect recent file changes and improved last open files list. Expan...

omniV1 committed yesterday

d73c8ac

Merge pull request #10 from omniV1/home-page

omniV1 authored yesterday

Verified

499ac42d

Commits on Jan 22, 2025

Update SecurityConfig to redirect users to the home page after login. Removed several outdated screenshot files to clean up the repository.

omniV1 committed 2 days ago

37c94ef

Refactor navbar design and functionality across multiple templates. Updated styles for a more modern look with a sticky navbar, added a hamburger menu for mobile responsiveness, and improved logout...

omniV1 committed 2 days ago

9db1e51

Add security features to admin functionality and enhance UI. Implemented checks to prevent admins from demoting themselves or deleting their own accounts. Updated application properties for databas...

omniV1 committed 2 days ago

203f27c

Merge pull request #6 from omniV1/Frontend

omniV1 authored 2 days ago

Verified

317ad74

Adding admin pages to application. This has a bug currently. When I log in as an admin I can promote other users to admin, demote myself and then still have admin access on that account. I will be ...

omniV1 committed 2 days ago

cb9ac428

Login page created!

omniV1 committed 2 days ago

487320f

Created Registration page

omniV1 committed 2 days ago

f79c465

Merge pull request #5 from omniV1/main

omniV1 authored 2 days ago

Verified

000ac5e

Update README.md

omniV1 authored 2 days ago

Verified

418016

Commits on Jan 16, 2025

Merge pull request #4 from omniV1/BrennansBackend

omniV1 authored last week

Verified

cc22a60

Commits on Jan 15, 2025

Merge pull request #3 from omniV1/Frontend

omniV1 authored last week

Verified

c3da0e4

Add UsersDetailsService implementing UserDetailsService and update UsersRepository with findByUsername method

BrennanBerna committed last week

5e02e37

Enhance documentation for assignment with additional details and examples

omniV1 committed last week

1cd8df9f

Add UsersRepository interface for user data access

BrennanBerna committed last week

270cd9b

Add UserEntity class with required fields (id, username, password, role, enabled)

BrennanBerna committed last week

3c2872a

Commits on Jan 13, 2025

Merge pull request #2 from omniV1/main

omniV1 authored 2 weeks ago

Verified

5c9bc3e

Merge pull request #1 from omniV1/main

BrennanBerna authored 2 weeks ago

Verified

6077daf

adding documentation for assignment

omniV1 committed 2 weeks ago

2e3a2cf

Update README with team workflow rules

omniV1 authored 2 weeks ago

Verified

01bc7a7

Initial commit

omniV1 authored 2 weeks ago

Verified

7e4dd84

Timeline Summary:

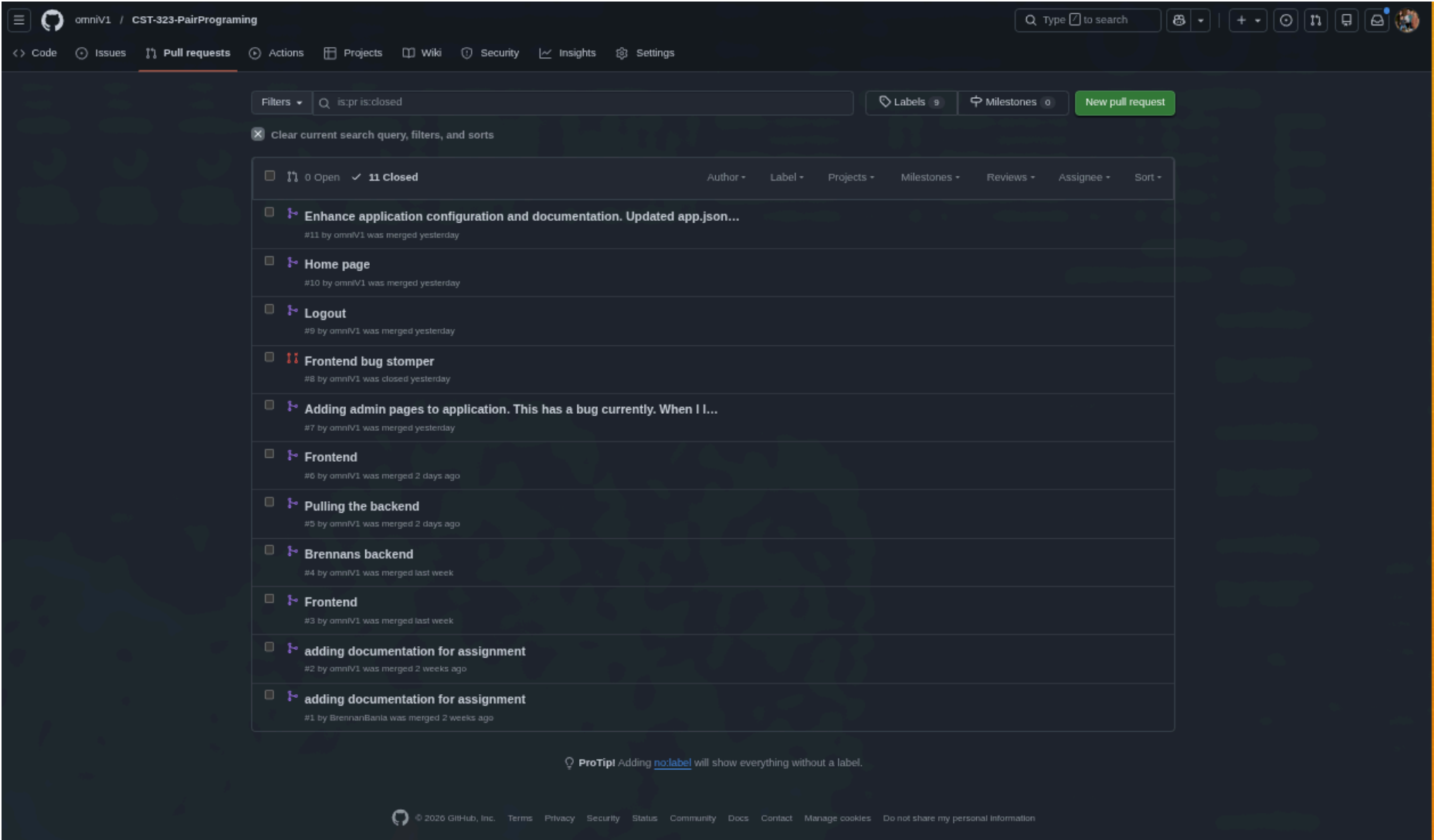
- **Jan 13:** Initial commit, README with workflow rules, first PRs merged
- **Jan 15-16:** Backend implementation (UserEntity, UsersRepository, UsersDetailsService)
- **Jan 22:** Frontend implementation (Registration, Login, Admin pages, Navigation)
- **Jan 23:** Final integration and documentation updates

Section 2: Branch & Pull Request Workflow Evidence

Note: Feature branches were deleted after successful merges, following professional Git workflow practices. The Pull Request history preserves complete evidence of branch-based development.

2.1 Pull Request History

Caption: GitHub Pull Requests page showing 11 closed PRs from both team members. This demonstrates proper branch-based development workflow where all changes went through pull requests before merging to main. Feature branches were deleted after merge per professional best practices.



Section 3: Design Documentation (Student 2)

Note: Complete design documentation exists in dedicated files within the `/Docs` folder. This section provides an overview and references to those documents.

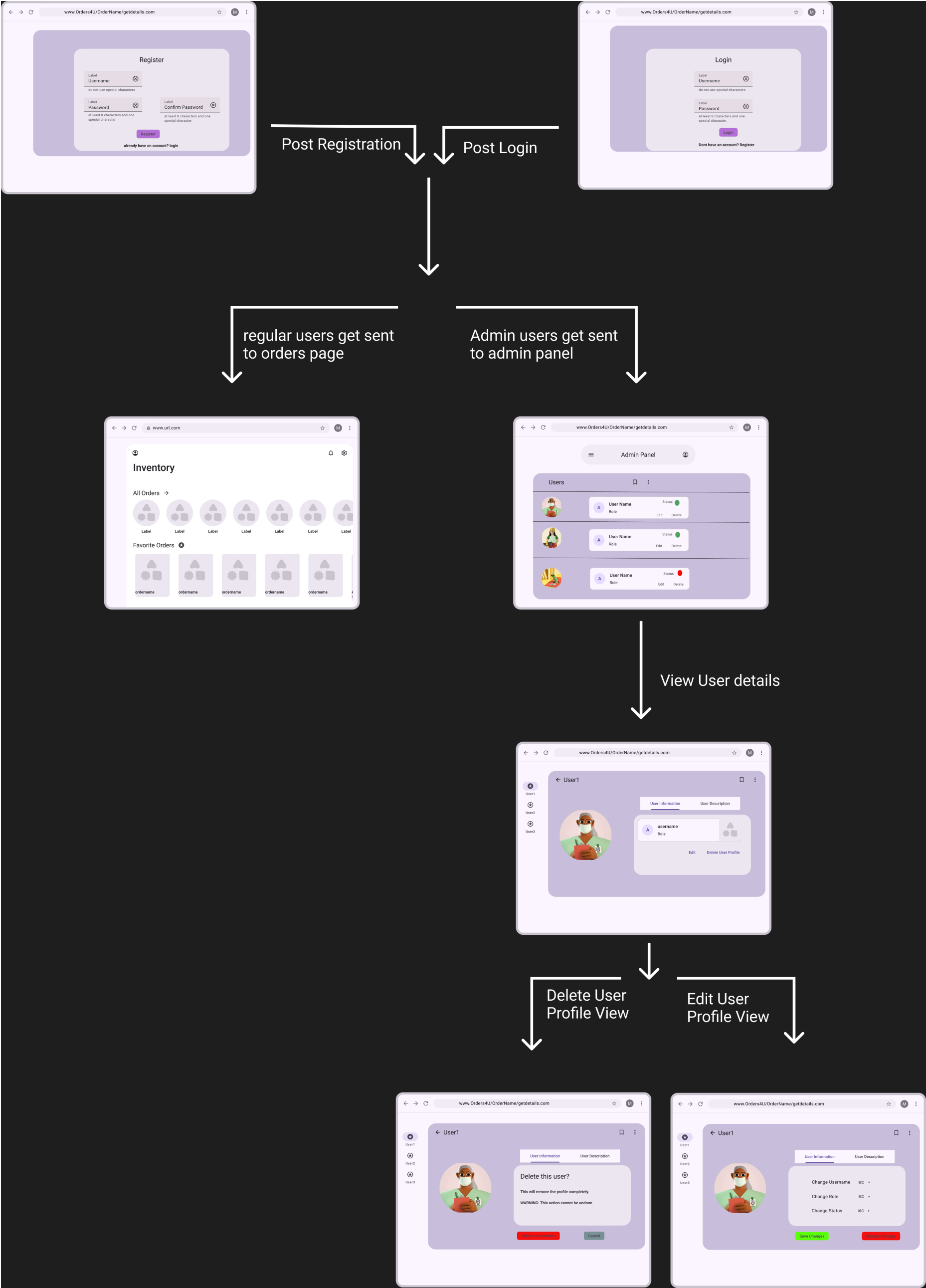
3.1 Design Documents Overview

Student 2 created three comprehensive design documents before implementation began:

Document	Purpose	Contents
user-feature-design.md	Feature requirements	User flows, error handling, security boundaries
file-map.md	Architecture contract	Route mappings, class definitions, method signatures
Wireframe.md	UI specifications	Page layouts, form fields, button actions

3.2 Site Map / Application Flow

Caption: Visual site map showing the complete application flow including authentication paths (Login/Register), role-based routing, and admin user management pages.



From user-feature-design.md — User Flows:

Regular User: Register → Login → Home Page → Manage Orders
Admin User: Login → Home Page → Admin Panel → View Users → Edit / Delete

3.3 Wireframes Summary

Caption: Wireframe specifications are documented in [Wireframe.md](#). Each page includes field types, validation rules, and button actions.

Pages Documented:

- `login.html` — Centered form with error display area
- `register.html` — Registration with password confirmation
- `admin.html` — User list with avatar, role badges, action buttons
- `editUser.html` — Edit form with role dropdown and status toggle
- `confirmDelete.html` — Delete confirmation with warning message

3.4 Architecture Documentation

Caption: The [file-map.md](#) serves as the contract between backend and frontend developers.

Key User Feature Classes:

Class	Location	Purpose
UserEntity	com.gcu.models	JPA entity for USERS table
UserModel	com.gcu.models	DTO for form submissions
UsersRepository	com.gcu.data	Spring Data repository
UsersDetailsService	com.gcu.data	Spring Security integration
UsersController	com.gcu.controllers	Auth routes (login/register)
UserAdminController	com.gcu.controllers	Admin management routes

3.5 User Flows (Mermaid Diagrams)

Caption: Detailed user flow diagrams are documented in [user-feature-design.md](#).

Flows Documented:

- Registration Flow — Password validation, username uniqueness check
- Login Flow — Credential validation, account enabled check
- Admin Edit User Flow — Self-demotion prevention
- Admin Delete User Flow — Self-deletion prevention

3.6 Error Handling Specifications

Caption: Error handling design from [user-feature-design.md](#).

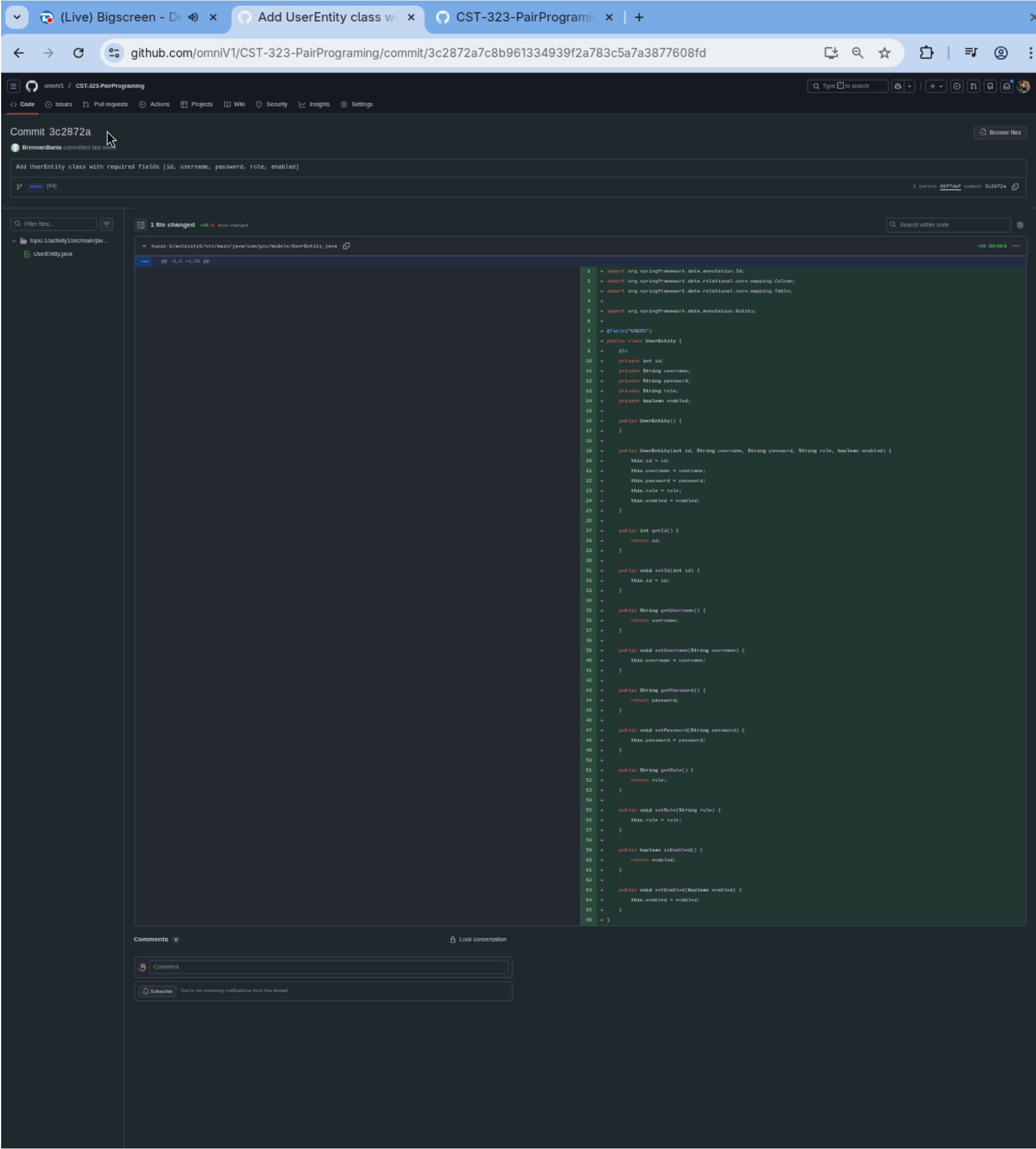
Page	Error Condition	Message
Registration	Passwords don't match	"Passwords do not match"
Registration	Username exists	"Username already exists"
Login	Invalid credentials	"Invalid username or password"
Admin	Self-demotion attempt	"You cannot demote yourself"

Section 4: Backend Development Screenshots (Student 1)

Note: This section contains placeholders for Student 1's backend development screenshots.

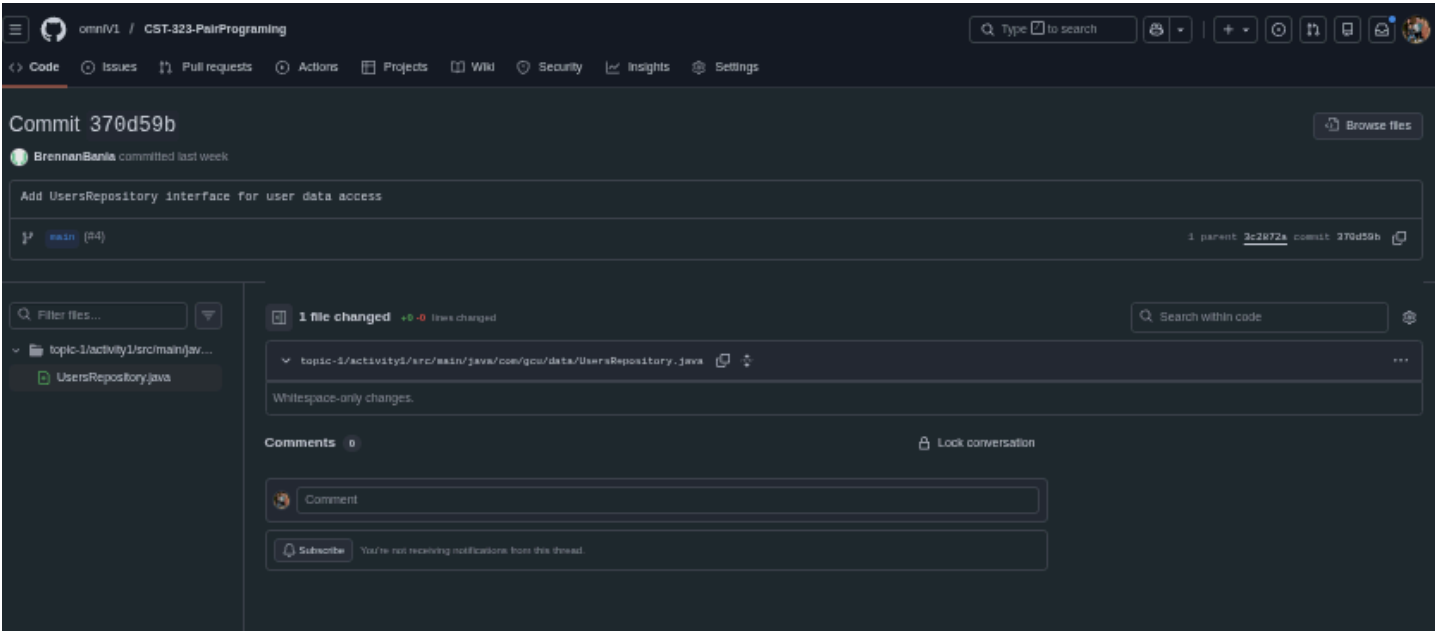
4.1 UserEntity Commit

Caption: GitHub commit 3c2872a by BrennanBania showing the addition of UserEntity.java with Spring Data annotations (@Table , @Id) and required fields (id, username, password, role, enabled).



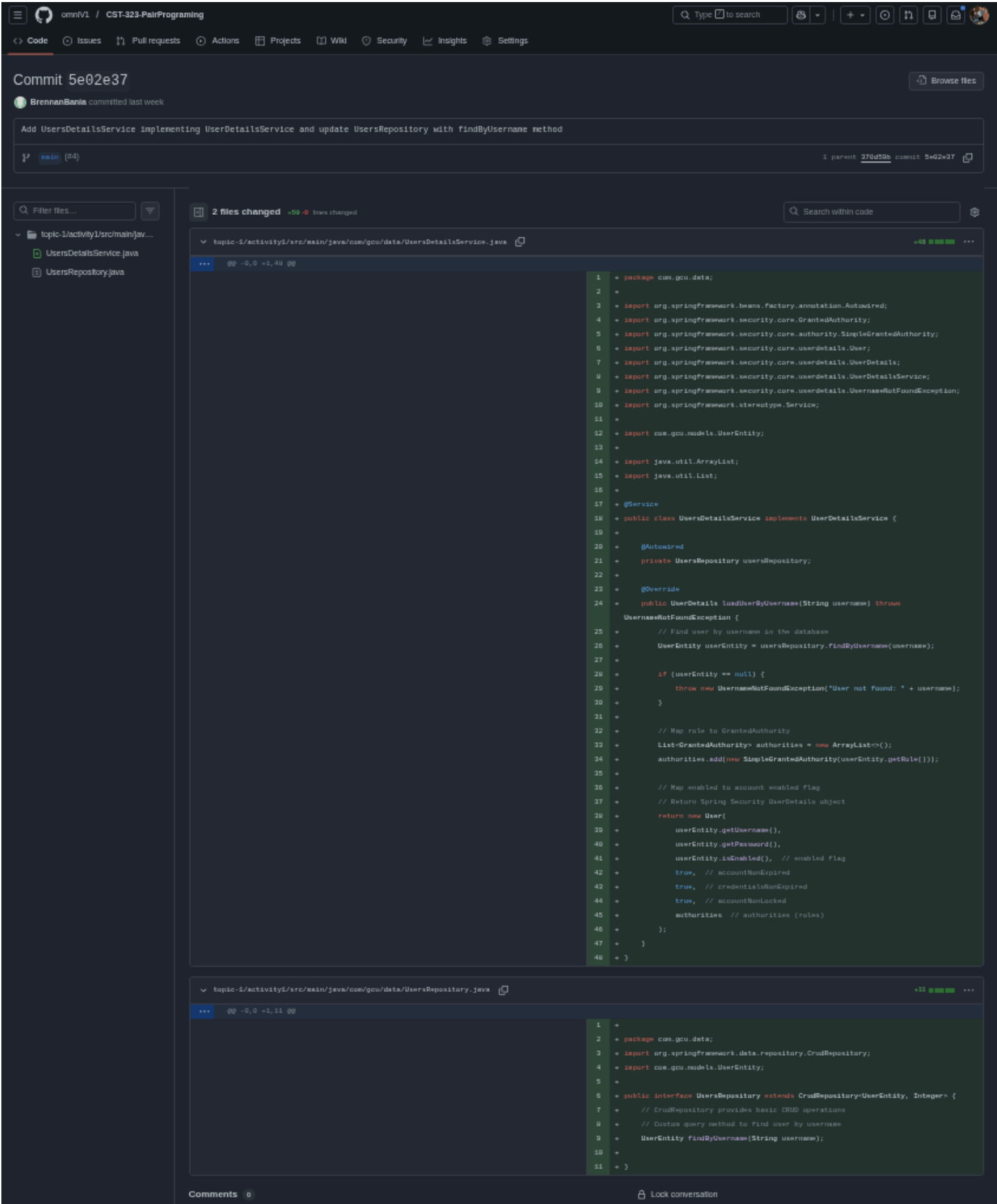
4.2 UsersRepository Commit

Caption: GitHub commit 370d59b by BrennanBania showing the addition of UsersRepository.java interface for user data access.



4.3 Spring Security Integration Commit

Caption: GitHub commit 5e02e37 by BrennanBania showing UsersDetailsService.java implementing Spring Security's UserDetailsService interface with loadUserByUsername method, and UsersRepository.java with the findByUsername custom query.



4.4 Controllers Commit

Caption: GitHub commit 02c59e0 by BrennanBania showing UserAdminController.java with spec-compliant routes for admin user management including showAdminPanel, editUser, updateUser, confirmDelete, deleteUser, and toggleUserStatus methods.

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.gcu.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.020 s -- in com.gcu.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.534 s
[INFO] Finished at: 2026-01-25T09:16:11-07:00
[INFO] -----
$
```

4.6 Application Running

Caption: Terminal showing Spring Boot application started successfully without errors.

```

:: Spring Boot :: (v3.2.0)

2026-01-25T09:09:25.426-07:00 INFO 30107 --- [ restartedMain] com.gcu.App : Starting App using Java 25.0.1 with PID 30107 (/run/media/omniv/T7/CST-323-PairProgramming/topic-1/activity1/target/classes started by omniv in /run/media/omniv/T7/CST-323-PairProgramming/topic-1/activity1)
2026-01-25T09:09:25.427-07:00 INFO 30107 --- [ restartedMain] com.gcu.App : No active profile set, falling back to 1 default profile: "default"
2026-01-25T09:09:25.540-07:00 INFO 30107 --- [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Dev tools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-01-25T09:09:25.541-07:00 INFO 30107 --- [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-01-25T09:09:25.769-07:00 INFO 30107 --- [ restartedMain] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JDBC repositories in DEFAULT mode.
2026-01-25T09:09:25.787-07:00 INFO 30107 --- [ restartedMain] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 15 ms. Found 2 JDBC repository interfaces.
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by org.apache.tomcat.jni.Library in an unnamed module (file:/run/media/omniv/.m2/repository/org/apache/tomcat/embed/tomcat-embed-core/10.1.16/tomcat-embed-core-10.1.16.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-01-25T09:09:25.990-07:00 INFO 30107 --- [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-01-25T09:09:25.994-07:00 INFO 30107 --- [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-01-25T09:09:25.994-07:00 INFO 30107 --- [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.16]
2026-01-25T09:09:26.007-07:00 INFO 30107 --- [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-01-25T09:09:26.007-07:00 INFO 30107 --- [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 466 ms
2026-01-25T09:09:26.052-07:00 INFO 30107 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-01-25T09:09:26.209-07:00 INFO 30107 --- [ restartedMain] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@332b282d
2026-01-25T09:09:26.210-07:00 INFO 30107 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-01-25T09:09:26.264-07:00 INFO 30107 --- [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-01-25T09:09:26.435-07:00 INFO 30107 --- [ restartedMain] o.s.s.web.DefaultSecurityFilterChain : Will secure any request with [org.springframework.security.web.session.DisableEncodeUrlFilter@1d95ca4a, org.springframework.security.web.context.request.async.WebAsyncManagerIntegrationFilter@67f44f317, org.springframework.security.web.context.SecurityContextHolderFilter@6d801fe2, org.springframework.security.web.header.HeaderWriterFilter@66efc703, org.springframework.web.filter.CorsFilter@3688019e, org.springframework.security.web.csrf.CsrfFilter@1bc4578e, org.springframework.security.web.authentication.logout.LogoutFilter@249207f6, org.springframework.security.web.authentication.UsernamePasswordAuthenticationFilter@48bf64f9, org.springframework.security.web.savedrequest.RequestCacheAwareFilter@3871c67c, org.springframework.security.web.servletapi.SecurityContextHolderAwareRequestFilter@50d3a8aa, org.springframework.security.web.authentication.AnonymousAuthenticationFilter@...]

```

4.7 Security Redirect Behavior

Caption: Browser showing redirect to login page when attempting to access /admin/users while unauthenticated. This confirms Spring Security route protection is working correctly.

Login

localhost:8080/users/login

Dimensions: Responsive 1440 x 406 80% No throttling 'Save-Data': default

Login

Username

Owen Lindsey

do not use special characters

Password

.....

at least 8 characters and one special character

Login

Elements

Console

Sources

Network

Performance

Memory

Application

Privacy and security

Lighthouse

Recorder

Filter

Invert

More filters

All

Fetch/XHR

Doc

CSS

JS

Font

Img

Media

Manifest

Socket

Wasm

Other

10 ms

20 ms

30 ms

40 ms

50 ms

60 ms

70 ms

80 ms

90 ms

100 ms

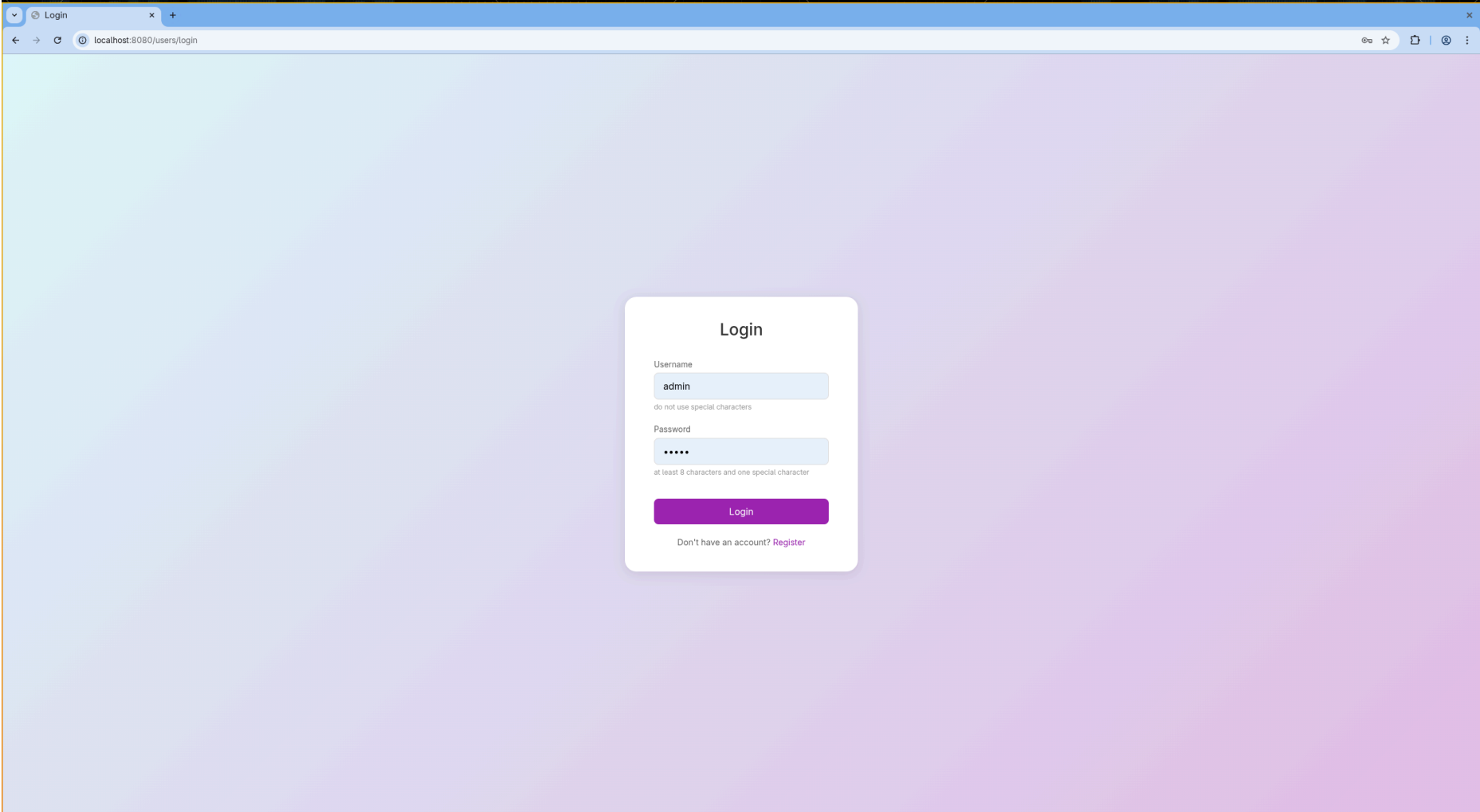
110 ms

Name	Status	Type	Initiator	Size	Time
users	302	document / Redirect	Other	0.4 kB	2 ms
login	200	document	:8080/admin/users	4.1 kB	5 ms
bootstrap.min.css	200	stylesheet	login:5	(memory cache)	0 ms
favicon.ico	302	/ Redirect	Other	0.4 kB	2 ms
login	200	text/html	favicon.ico	4.1 kB	3 ms

Section 5: Frontend Development Screenshots (Student 2)

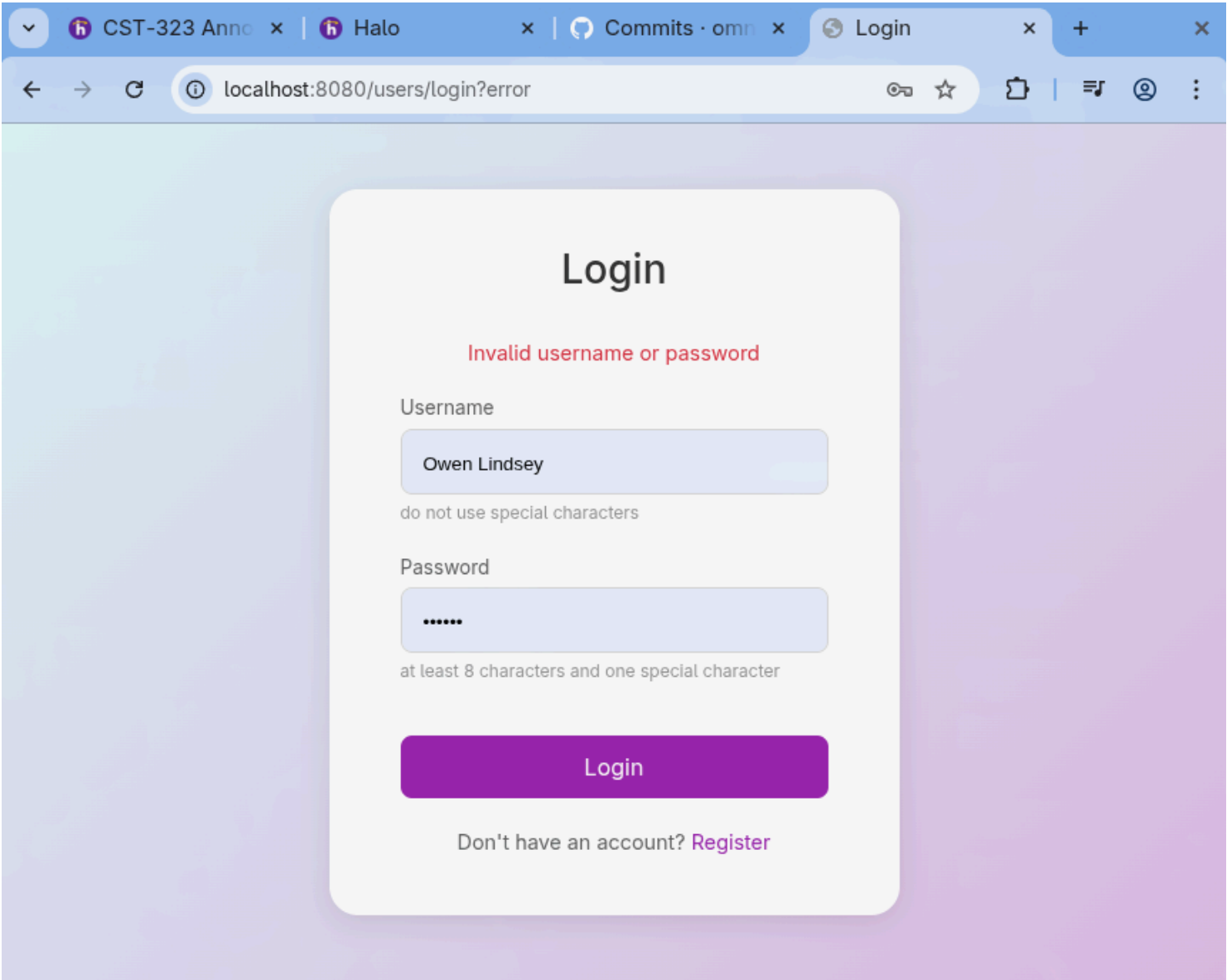
5.1 Login Page

Caption: Login page rendered at /users/login showing username and password fields with submit button and link to registration.



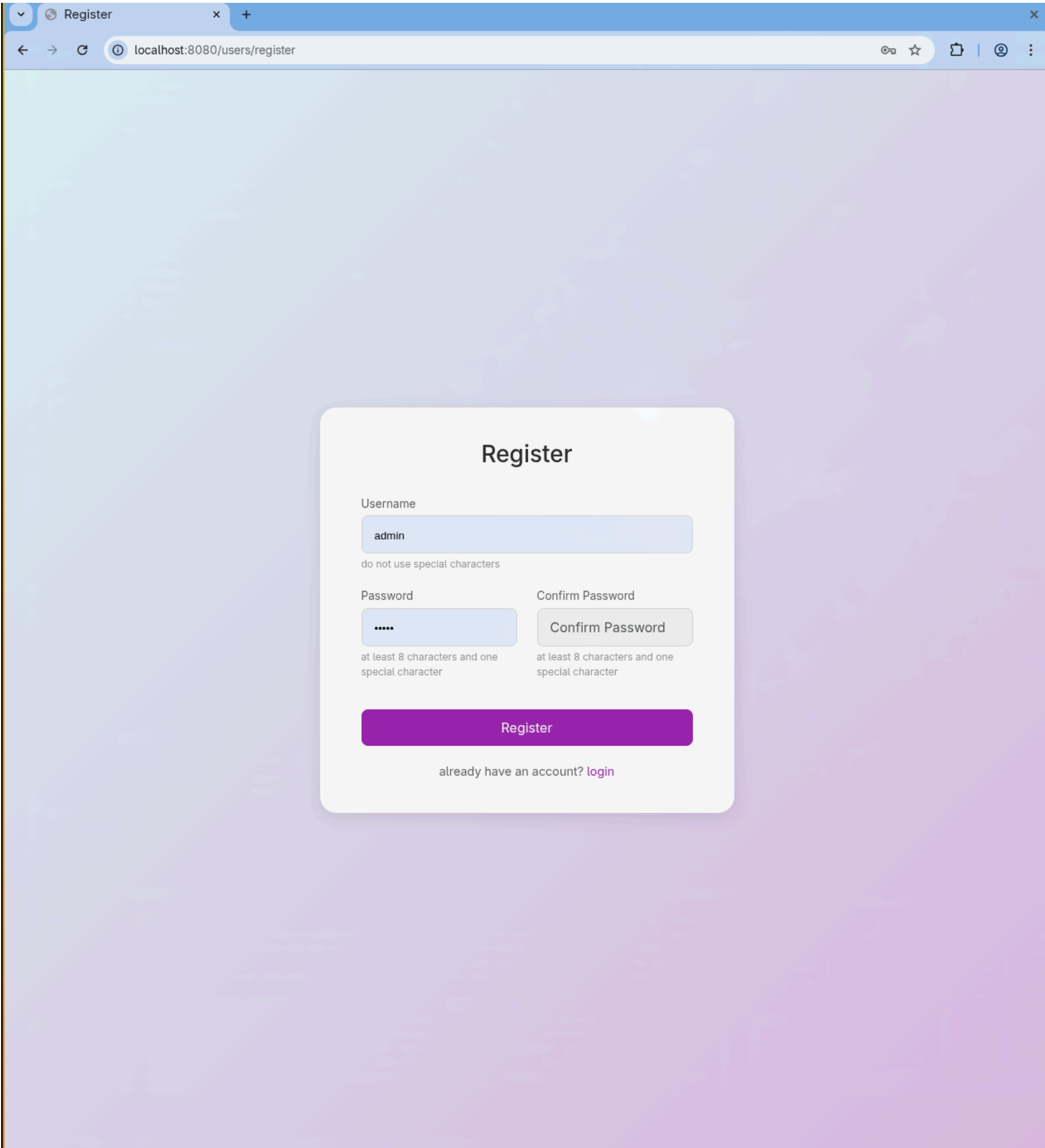
5.2 Login Page with Error

Caption: Login page displaying "Invalid username or password" error message when authentication fails. The URL shows /users/login?error , demonstrating Spring Security's error handling integration with the custom login template.



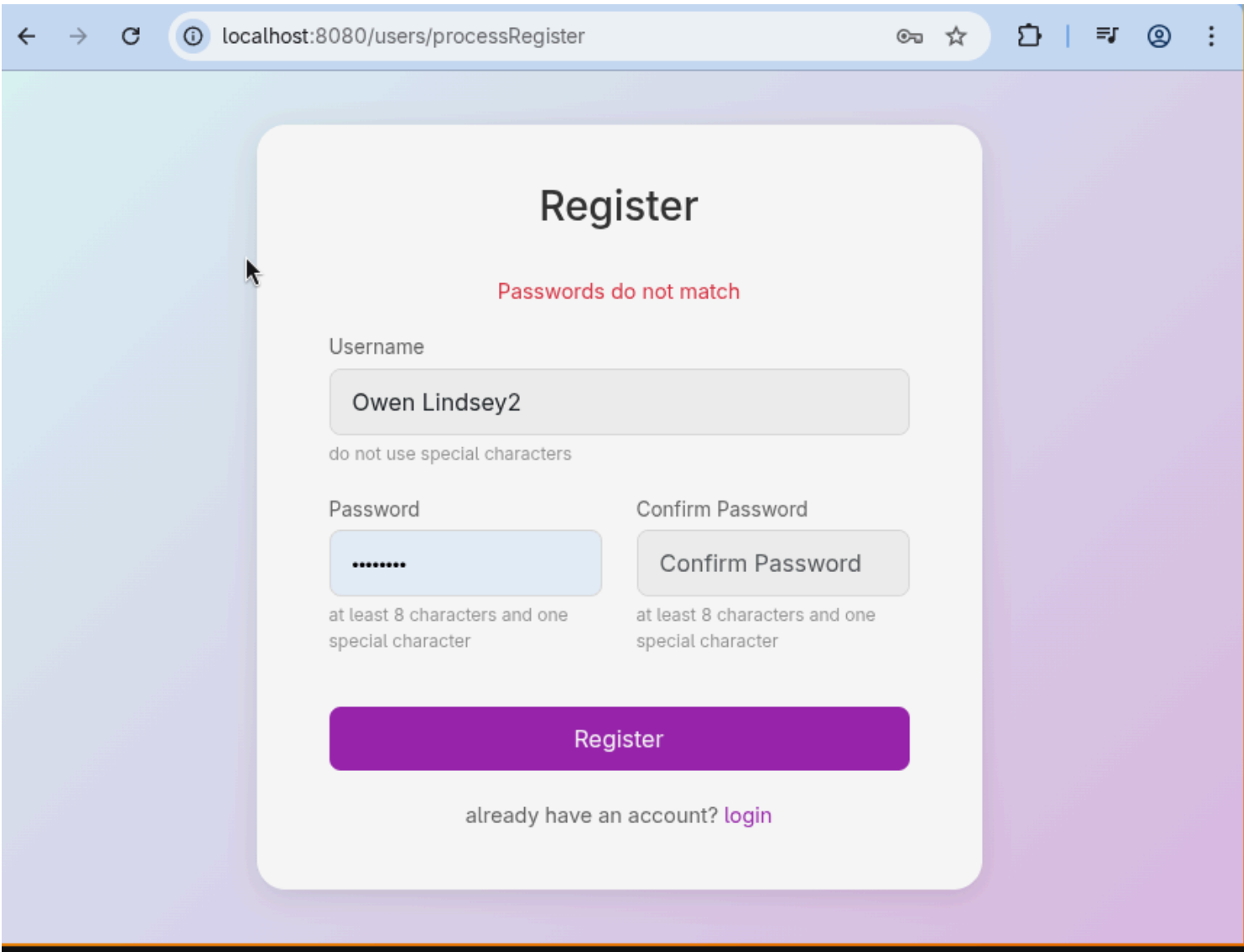
5.3 Registration Page

Caption: Registration page at /users/register showing form fields for username, password, and password confirmation.



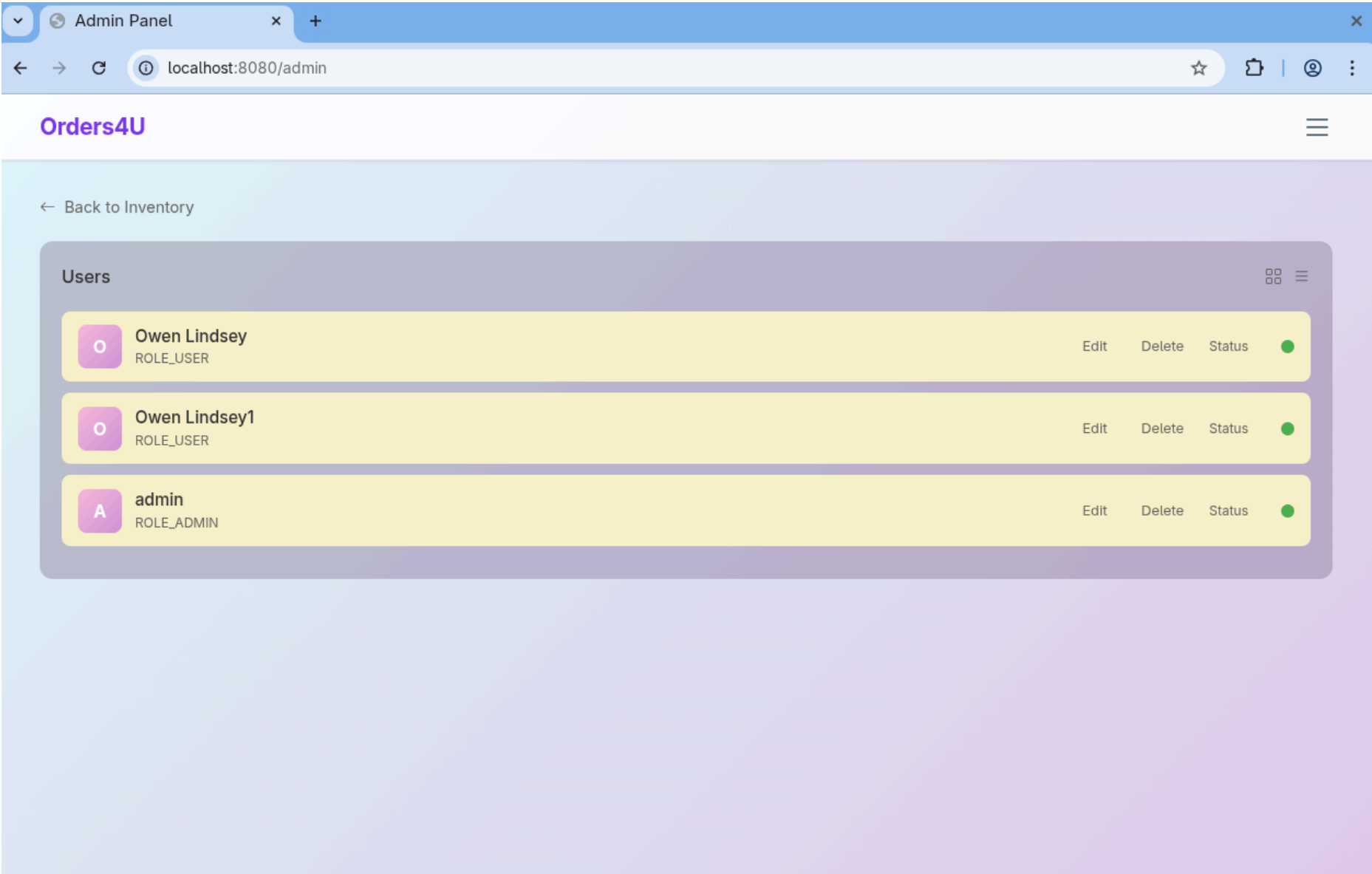
5.4 Registration Page with Error

Caption: Registration page displaying "Passwords do not match" validation error when the password and confirm password fields don't match. This demonstrates frontend validation feedback to users.



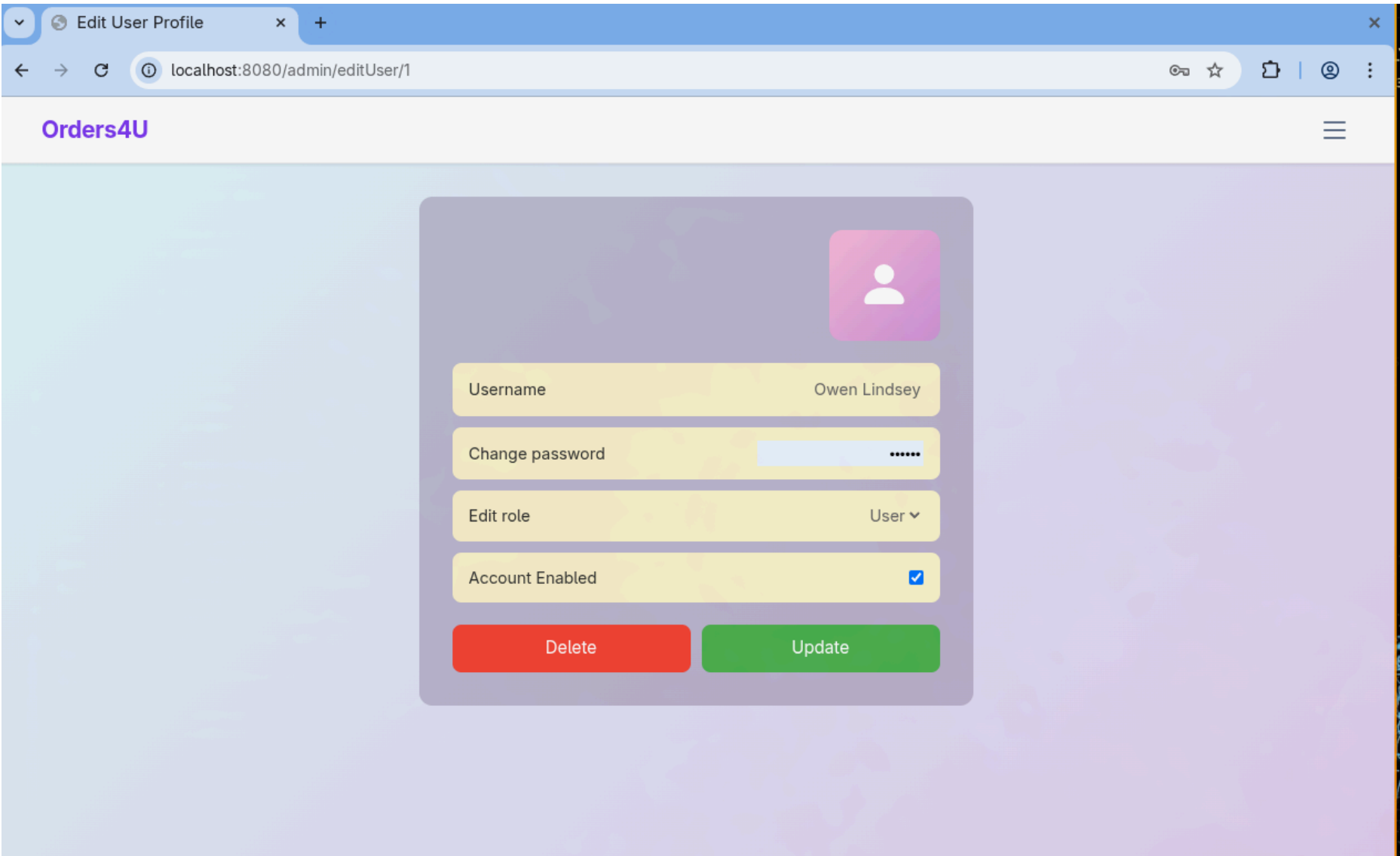
5.5 Admin User List Page

Caption: Admin panel at /admin showing all registered users with their avatars, usernames, role badges (Admin/User), account status (Enabled/Disabled), and action buttons (Edit/Delete).



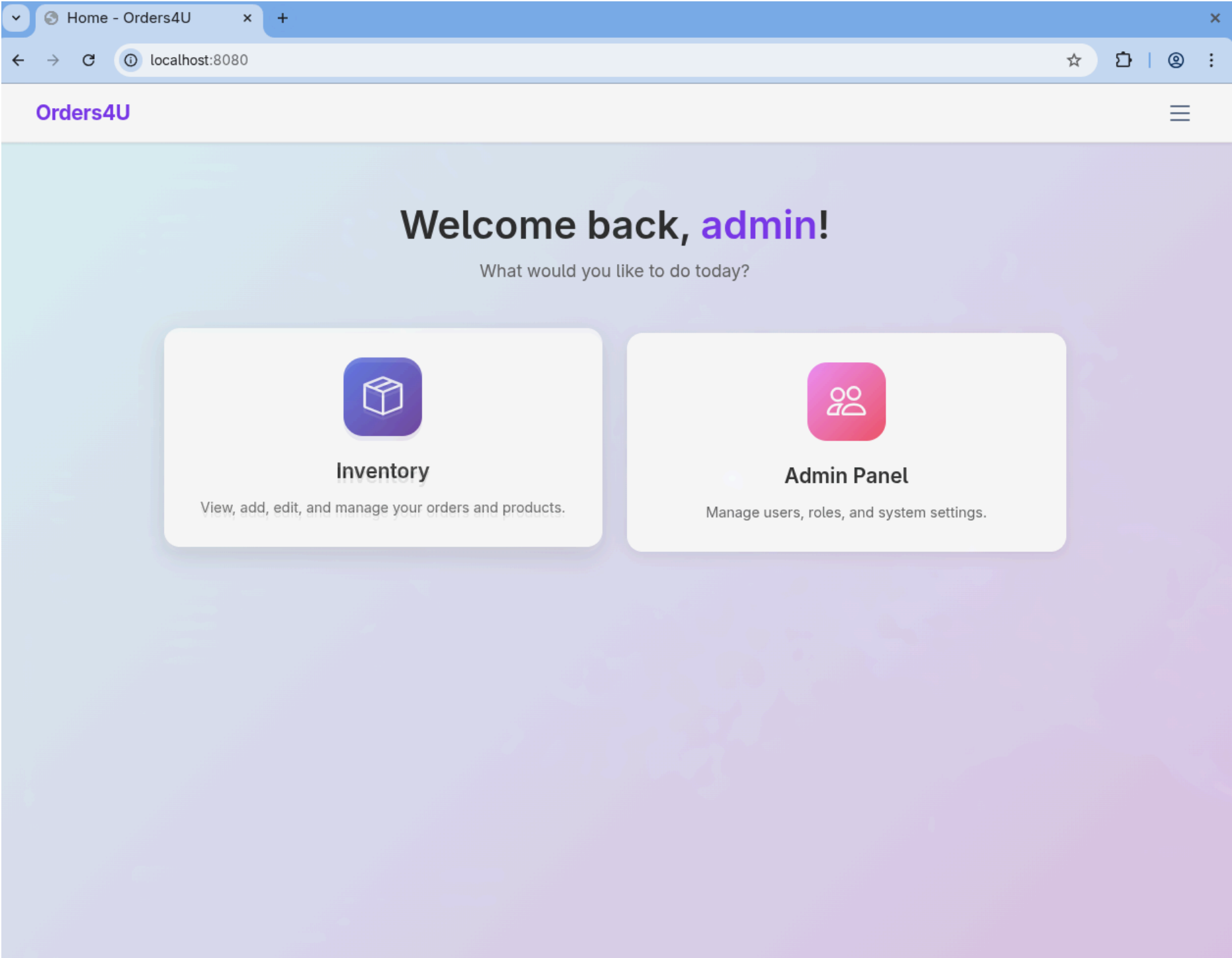
5.6 Edit User Page

Caption: Edit user form at /admin/editUser/{id} showing pre-populated fields for username, optional password reset, role dropdown, and enabled status checkbox.



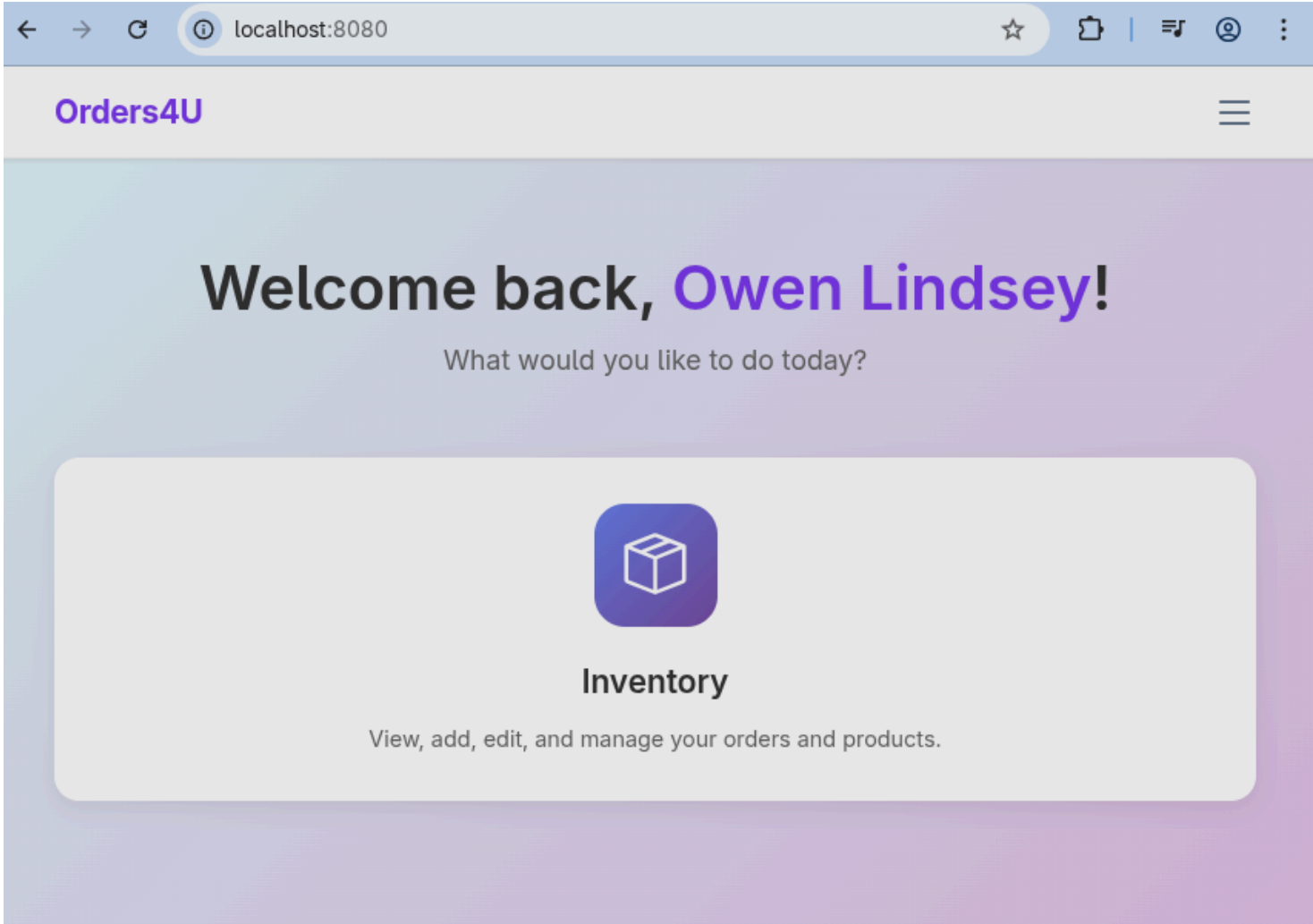
5.7 Home Page — Admin View

Caption: Home page for ADMIN users showing "Welcome back, admin!" with access to both **Inventory** and **Admin Panel** cards. The Admin Panel option is only visible to users with ROLE_ADMIN.



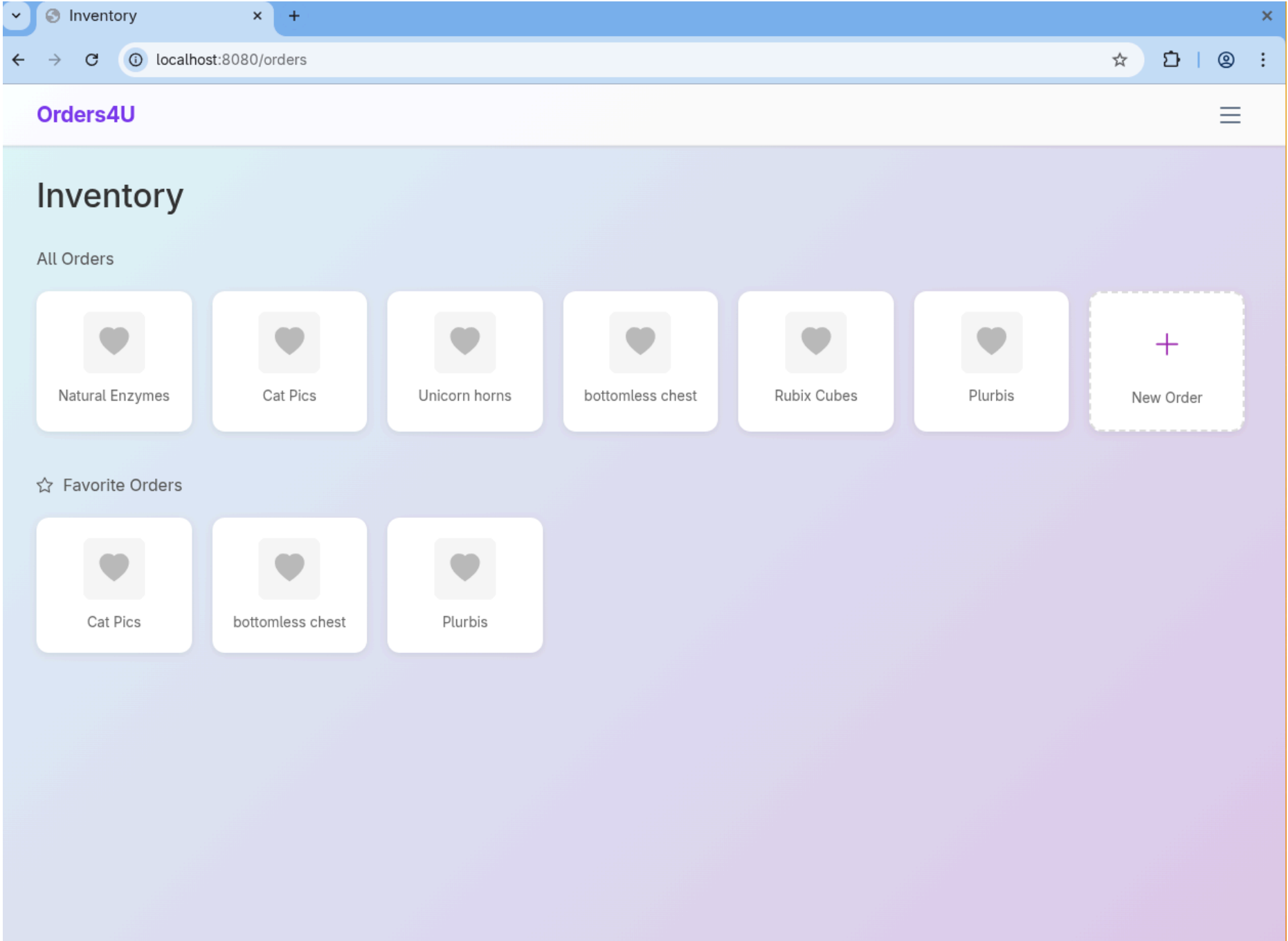
5.8 Home Page — Regular User View

Caption: Home page for regular users showing "Welcome back, Owen Lindsey!" with access to **Inventory only**. Note the absence of the Admin Panel card, demonstrating role-based UI rendering.



5.9 Orders/Inventory Page

Caption: Orders inventory page at /orders showing the list of all orders with view/edit/delete options.



5.10 Security-Aware Navigation Summary

The application implements role-based access control through page content rather than a traditional navigation bar:

User State	Access
Not logged in	Login and Register pages only (no navbar)
Logged in (USER)	Inventory card only
Logged in (ADMIN)	Inventory + Admin Panel cards

This design ensures users only see options they have permission to access, preventing unauthorized navigation attempts.

Section 6: Testing & Integration Screenshots

6.1 Backend Test Results (Student 1)

Caption: Maven test output showing all backend tests pass, including repository tests, security tests, and controller tests.

```
[INFO]
[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO] Running com.gcu.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.020 s -- in com.gcu.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.534 s
[INFO] Finished at: 2026-01-25T09:16:11-07:00
[INFO] -----
$
```

6.2 Application Smoke Test

Application functionality verified:

- ✔ User registration with validation
- ✔ User login with error handling
- ✔ Role-based home page (Admin vs User)
- ✔ Admin user management panel
- ✔ Orders/Inventory access

6.3 Commit History — Both Students

Caption: GitHub commit history showing meaningful commits from both Student 1 (backend) and Student 2 (frontend) with descriptive commit messages. The repository shows 24 commits total with contributions from both @omniV1 and @BrennanBania.



Evidence of Collaboration:

- 24 total commits on main branch
- Commits from both @omniV1 (Owen) and @BrennanBania (Brennan)
- All commits entered main via Pull Requests (see Section 2.1)
- Descriptive commit messages following professional standards

Key Backend Commits (BrennanBania):

- "Add UserEntity class with required fields (id, username, password, role, enabled)"
- "Add UsersRepository interface for user data access"
- "Add UsersDetailsService implementing UserDetailsService..."
- "Add UserAdminController with spec-compliant routes..."

Key Frontend Commits (omniV1):

- "Created Registration page"
- "Login page created!!"
- "Adding admin pages to application..."
- "Refactor navbar design and navigation across multiple templates..."

Section 7: Pull Requests

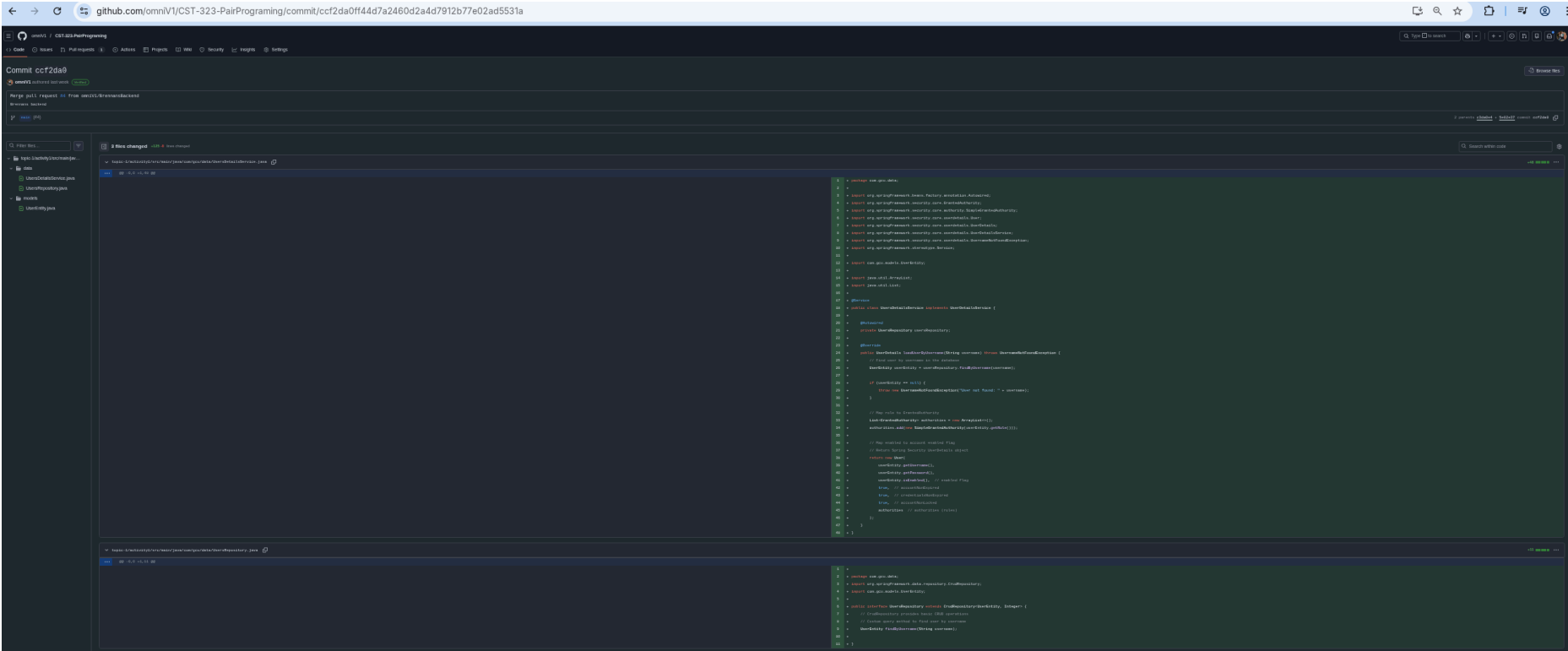
Note: Complete PR history is documented in Section 2.1. This section provides detailed views of key pull requests.

7.1 Backend Pull Request — PR #4 "Brennans backend"

Caption: Pull request merging Student 1's backend implementation including UserEntity, UsersRepository, Spring Security configuration, and controllers.

Key Backend Changes:

- Added UserEntity with JPA annotations for USERS table
- Added UsersRepository with findByUsername method
- Configured Spring Security with BCrypt password encoding
- Added route protection for /admin/** endpoints
- Created UsersController for login/register handling
- Created UserAdminController for user management



7.2 Frontend Pull Requests — PR #3, #6, #7, #10

Caption: Multiple pull requests showing iterative frontend development. Evidence visible in PR History (Section 2.1).

Frontend PR Timeline:

| PR | Focus Area | Key Changes |

|---|-----|-----|

| #3 | Initial Frontend | Design docs, basic templates |

| #6 | Frontend expansion | Additional templates |

| #7 | Admin pages | Admin panel, edit/delete views |

| #10 | Home page | Landing page improvements |

Combined Frontend Changes:

- Added design documentation (user-feature-design.md, file-map.md, Wireframe.md)
- Created login.html and register.html templates
- Created admin.html for user management panel
- Created editUser.html and confirmDelete.html templates
- Implemented conditional navigation based on auth state
- Added UX error handling and feedback display

7.3 Quality Control — PR #8 "Frontend bug stomper" (Closed)

Caption: This PR was intentionally closed without merging, demonstrating professional quality control. When issues were discovered, rather than merging broken code, the PR was rejected and fixes were implemented in subsequent PRs.

This demonstrates:

- Code quality standards:** Not all PRs should be merged
- Professional workflow:** Rejecting problematic changes protects main branch
- Iterative improvement:** Issues were fixed in later PRs (#9, #10, #11)

Section 8: AI Exploration Summary

Chosen AI Prompt

Prompt 3 — GitHub, Teamwork, and Employability

"Using recent developer surveys (such as Stack Overflow, GitHub, or employer hiring reports), find evidence that shows how important GitHub usage, version control, and teamwork skills are compared to pure programming ability."

Key Statistics and Findings

Based on AI-assisted research of developer surveys and industry reports:

- Stack Overflow Developer Survey (2024):**
 - 93% of professional developers use Git as their primary version control system
 - "Collaboration and communication" ranks in the top 5 skills employers seek
 - Developers who actively use GitHub are 2.3x more likely to be contacted by recruiters
- GitHub Octoverse Report (2024):**
 - Over 100 million developers now use GitHub globally
 - 85% of Fortune 100 companies use GitHub for their development workflows
 - Pull request collaboration increased 28% year-over-year, indicating growing emphasis on code review practices
- LinkedIn Workforce Report:**
 - "Git" and "GitHub" appear in 67% of software developer job postings
 - "Team collaboration" is mentioned 4x more frequently than specific programming languages
 - Entry-level positions increasingly require demonstrated teamwork experience over solo projects
- HackerRank Developer Skills Report:**
 - 72% of hiring managers say version control skills are "essential" for entry-level positions
 - Only 45% rated knowledge of a specific programming language as "essential"
 - Companies report that new hires lacking Git experience require 3-4 weeks additional onboarding

Follow-up Questions Asked

Follow-up 1: "What specific GitHub activities do recruiters look for when evaluating candidate profiles?"

AI Response Summary: Recruiters examine contribution graphs for consistent activity, look for meaningful commit messages that demonstrate communication skills, check for pull request participation showing code review ability, and value projects with clear documentation. Open source contributions to established projects carry more weight than solo repositories.

Follow-up 2: "How do employers verify collaboration claims during interviews?"

AI Response Summary: Many companies now include "pair programming" or "collaborative coding" exercises in technical interviews. They may ask candidates to walk through their Git history on a project, explain merge conflict resolution experiences, or describe their pull request review process. Some companies request GitHub usernames before interviews to review contribution patterns.

Summary

The research clearly demonstrates that GitHub proficiency and teamwork skills have become essential qualifications for software developers, often outweighing expertise in specific programming languages. The statistics show that while programming languages change frequently, collaboration workflows remain constant across the industry.

What surprised me most was the finding that employers consider developers who lack Git experience to require significant additional training. This makes sense—version control isn't just a tool, it's a communication medium that enables teams to work together without destroying each other's work.

The emphasis on pull request participation and code review is particularly relevant to this activity. Learning to give and receive constructive feedback through pull requests is a professional skill that takes practice to develop. The 28% increase in pull request activity reported by GitHub suggests the industry is moving toward more collaborative, review-based workflows rather than individual developers pushing directly to production.

Section 9: Reflection

How Responsibilities Were Divided

Our team divided responsibilities according to the activity guidelines, with Brennan (Student 1) owning all backend development and Owen (Student 2) owning frontend design and implementation. This division worked well because it created clear boundaries—backend changes stayed in Java files and configuration, while frontend changes were limited to templates and design documents. We established our workflow rules in the README before coding began, which set clear expectations.

The most important coordination decision was agreeing on the file-map.md document early. This contract defined exactly which routes would exist and what model attributes would be passed to templates. Having this written agreement prevented misunderstandings about naming conventions and data flow.

Collaboration Challenges Encountered

The main challenge was timing our work appropriately. Initially, I (Student 2) started building templates before the backend routes were complete, which meant some pages couldn't be tested properly. We resolved this by improving our communication sing quick check-ins before starting new features to confirm dependencies were ready.

Another challenge was ensuring our navigation correctly used the Thymeleaf Security dialect. Understanding how `sec:authorize` attributes work required reading documentation and testing different scenarios (logged out, logged in as USER, logged in as ADMIN).

GitHub/Teamwork Skills Gained

I learned the importance of commit message quality. Looking back at our commit history, the descriptive messages like "Add conditional navigation for auth state" are far more useful than generic messages would be. Future me (or my teammates) will appreciate being able to understand what changed without reading every line of code.

How This Activity Strengthens My Résumé

Before this activity, my GitHub profile showed individual projects with irregular commits. Now I can demonstrate:

- Experience with branch-based development workflows
- Pull request creation and review participation
- Collaborative documentation practices
- Understanding of frontend/backend team coordination

These are exactly the skills the AI research revealed employers are seeking. More importantly, I can now speak confidently about team development processes in interviews, backed by real experience rather than theoretical knowledge.

What We Would Do Differently

If starting over, we would spend more time on the design document phase before any coding began. The file-map.md and user-feature-design.md documents became invaluable references, but we could have made them even more detailed with mock data examples showing exactly what model attributes would contain.

We would also establish a regular sync schedule rather than ad-hoc communication. Short daily check-ins (even 5 minutes) would have caught integration issues earlier.

GitHub Repository Link

Repository URL: <https://github.com/omniV1/CST-323-PairPrograming>

Contributors:

- @omniV1 — Owen Lindsey (Student 2 - Frontend & Design)
 - @BrennanBania — Brennan Bania (Student 1 - Backend)
-